

# Problem Statement

## Product Dissection for top leading Platforms

Welcome to this case study on dissecting and designing products for top-leading platforms. In this case study, you will delve into the intriguing world of schema design for a prominent platform of your choice. Your task is to choose a top leading platform, research its features, and meticulously craft a schema design that encapsulates the essence of its functionality. By focusing on key entities, attributes, and relationships, you will gain invaluable insights into how data architecture drives the platform's effectiveness.

### Step 1: Choose a Leading Platform

Select a leading platform of your choice, which could span various domains such as social media, e-commerce, finance, or any other industry. This choice will form the foundation of your exploration into its schema design.

### Step 2: Research:

Thoroughly research the platform you have selected. Investigate its core features, functionalities, and user interactions. Identify the top features that define its user experience and contribute significantly to its popularity.

### Step 3: Product Dissection and Real-World Problems solved by the platform

In this step, you will meticulously analyse the platform's standout features and how they provide innovative solutions to real-world challenges. By identifying key functionalities that resonate with users, you'll unravel how the platform effectively addresses problems and enhances user experiences. This dissection will serve as the foundation for understanding how the schema design aligns with the platform's core objectives.

### Step 4: Case Study on the real-world problems and approach to solving them

In this pivotal step, you will expand on the real-world challenges uncovered in Step 3 through a comprehensive case study. Delve into specific instances where users encountered difficulties and showcase how the platform's unique features provided effective solutions. By dissecting the approach taken by the platform to overcome these challenges, you'll gain a deeper appreciation for the platform's user-centric design philosophy and how it shapes the schema design.

### Step 5: Schema Design Based on Top Features

Based on the features you have identified, craft a schema design that reflects the platform's data structure. Focus on the key entities, attributes, and relationships that underpin the chosen features. Your schema should capture the essence of how the platform organises and utilises its data.

### **Step 6: Rationale Behind the Design**

While creating the schema design, consider the rationale behind the platform's choices. Reflect on why certain entities and relationships were chosen and how they align with the platform's goals. This will help you understand the strategic decisions driving the schema's architecture.

### **Step 7: Create an ER Diagram**

Utilise tools like the Miro platform or similar applications to create an illustrative Entity-Relationship (ER) diagram. This diagram should vividly depict the entities, attributes, and relationships present within your schema design. The ER diagram will serve as a visual representation of your insights.

### **Step 8: Presentation of Findings**

Present your findings in a clear and concise manner. Showcase your understanding of how the schema design impacts the platform's functionality and user experience. Explain how your chosen features are integrated into the schema and how the schema's structure supports the platform's objectives.

### **Task Details:**

1. **Answer Submission:** Your submission should include well-structured solutions for all provided questions related to product schema designs.
2. **Video Creation:** Create an informative and engaging video where you thoroughly explain the case study.
3. **Depth and Clarity:** Ensure your solutions are detailed and showcase your understanding of product schema design principles. Similarly, in the video, provide clear explanations that are easy to understand for a wide audience.
4. **Creativity Encouraged:** You are welcome to utilise visuals, diagrams, or creative elements to enhance the clarity and impact of your explanations.

### **Note:**

1. Duplicate this document and proceed to write your solutions and prepare your video.
2. Include the video link in this document before final submission.

Best of luck in completing this project and showcasing your prowess in dissecting and designing product schema for leading platforms! **For reference, we have also conducted a case study on Instagram, which you can find below. This case study will provide you with valuable insights into how schema design plays a pivotal role in shaping the functionality and success of a prominent platform.**



## Product Dissection for Amazon

### Company Overview:

Amazon is one of the world's largest e-commerce and technology companies, founded by **Jeff Bezos** in **1994**. Initially launched as an online bookstore, Amazon has expanded into a global marketplace offering millions of products across categories such as electronics, fashion, groceries, home appliances, books, healthcare, and digital services. Today, Amazon operates in numerous countries and serves millions of customers through its website and mobile application.

Amazon's mission is **"to be Earth's most customer-centric company,"** where customers can discover, purchase, and receive products quickly, securely, and conveniently. The company continuously improves customer experience by using technologies such as Artificial Intelligence (AI), Machine Learning (ML), cloud computing, and big data analytics.

Apart from serving customers, Amazon also provides opportunities for millions of sellers to reach a global audience through its marketplace model. This ecosystem benefits customers, sellers, delivery partners, and manufacturers by creating a seamless digital commerce experience.

### Product Dissection and Real-World Problems Solved by Amazon

Amazon has transformed the way people purchase products by solving several real-world challenges associated with traditional shopping. Before e-commerce platforms became popular, customers had to visit multiple physical stores, compare prices manually, and often

faced limited product availability. Amazon eliminated these difficulties by bringing millions of products together on a single platform.

One of Amazon's greatest strengths is its **advanced product search and recommendation system**. Customers can quickly find products using keywords, filters, categories, brands, price ranges, customer ratings, and recommendations based on previous purchases. This significantly reduces the time required to discover suitable products.

Amazon also addresses the issue of **trust in online shopping**. Every product includes detailed descriptions, high-quality images, customer ratings, verified reviews, seller information, warranty details, and frequently asked questions. These features help customers make informed purchasing decisions and reduce uncertainty before buying.

Another major challenge Amazon solves is **fast and reliable product delivery**. Through its extensive network of fulfillment centers, warehouses, and logistics partners, Amazon offers same-day, one-day, and scheduled delivery services in many locations. Customers can monitor every stage of their shipment through real-time order tracking.

Amazon has also simplified **online payment security** by offering multiple payment options, including credit/debit cards, UPI, net banking, digital wallets, EMI, Cash on Delivery (COD), and gift cards. Secure payment gateways and fraud detection systems help protect customer transactions.

For **sellers**, Amazon provides inventory management, order processing, advertising tools, sales analytics, and fulfillment services. These capabilities enable businesses of all sizes to expand their customer base without investing heavily in their own online infrastructure.

In addition, Amazon's **easy return and refund policy** reduces customer risk by allowing eligible products to be returned or replaced with minimal effort. This customer-centric approach builds long-term trust and encourages repeat purchases.

Overall, Amazon successfully solves problems related to **product discovery, price comparison, secure payments, logistics, inventory management, customer trust**, and efficient online shopping. Its technology-driven ecosystem continues to improve customer satisfaction while supporting millions of businesses worldwide.

## Case Study: Real-World Problems and Amazon's Innovative Solutions

Amazon has become a global leader in e-commerce by identifying common challenges faced by both customers and sellers and developing innovative technological solutions. The platform combines artificial intelligence, cloud computing, logistics, and data analytics to provide a seamless shopping experience.

## **Problem 1: Difficulty in Finding the Right Product**

### **Real-World Challenge**

Customers often spend significant time searching multiple stores to compare products, prices, brands, and specifications. This process can be time-consuming and confusing, especially when there are thousands of similar products available.

### **Amazon's Solution**

Amazon provides an intelligent product search engine supported by advanced filtering options such as category, brand, price, customer ratings, discounts, Prime eligibility, and delivery date. Machine learning algorithms analyze user behavior and recommend products based on browsing history, previous purchases, and customer preferences. This personalized shopping experience allows customers to discover relevant products quickly and efficiently.

## **Problem 2: Lack of Trust in Online Shopping**

### **Real-World Challenge**

Many customers hesitate to purchase products online because they cannot physically inspect the items. Concerns about product quality, counterfeit products, and unreliable sellers reduce consumer confidence.

### **Amazon's Solution**

Amazon improves transparency by providing verified customer reviews, product ratings, detailed descriptions, multiple product images, seller information, warranty details, and "Amazon's Choice" recommendations. These features help customers evaluate products before making purchasing decisions, increasing trust and reducing purchase risk.

## **Problem 3: Slow Delivery and Poor Order Visibility**

### **Real-World Challenge**

Traditional online shopping often involved long delivery times with limited information about shipment status, leading to customer frustration.

### **Amazon's Solution**

Amazon operates a sophisticated logistics network consisting of fulfillment centers, warehouses, delivery stations, and transportation partners. Customers receive estimated delivery dates, live shipment tracking, delivery notifications, and proof of delivery. Services such as Prime, Same-Day Delivery, and One-Day Delivery significantly improve customer satisfaction.

## **Problem 4: Complicated Return and Refund Process**

### **Real-World Challenge**

Many online retailers make product returns difficult, causing customer dissatisfaction and reducing confidence in online shopping.

### **Amazon's Solution**

Amazon offers a simplified return and refund system. Customers can initiate returns through their accounts, schedule product pickups, track refund status, and receive replacements or refunds according to product eligibility. This hassle-free process strengthens customer loyalty and encourages repeat purchases.

## **Problem 5: Challenges Faced by Small Businesses**

### **Real-World Challenge**

Small businesses often struggle to reach a large customer base due to limited marketing budgets, logistics infrastructure, and technical resources.

### **Amazon's Solution**

Amazon Marketplace enables businesses of all sizes to sell products online. Sellers receive access to inventory management tools, secure payment processing, advertising services, customer analytics, and Fulfillment by Amazon (FBA), where Amazon stores, packs, ships, and handles customer service on behalf of sellers. This allows businesses to scale efficiently while focusing on product development.

## **Conclusion**

Amazon's continuous innovation has transformed the global retail industry by addressing key challenges faced by customers and businesses. Through intelligent product recommendations, secure payment systems, reliable logistics, transparent reviews, and efficient seller support, Amazon delivers a customer-centric shopping experience. Its

technology-driven platform demonstrates how effective product design and data management can solve real-world problems while creating value for millions of users worldwide.

## **Top Features of Amazon**

### **1. Customer Account Management**

Amazon allows users to create personal accounts where they can securely manage their profile information, addresses, payment methods, wishlists, orders, and account settings. Each customer has a personalized dashboard that stores shopping history and preferences.

#### **Key Benefits**

- Secure login and authentication
- Multiple shipping addresses
- Saved payment methods
- Personalized shopping experience
- Easy order management

### **2. Product Catalog**

The Product Catalog is the heart of Amazon's marketplace. Millions of products are organized into categories and subcategories, allowing customers to browse and compare products easily.

Each product page includes:

- Product Name
- Brand
- Description
- Specifications
- Images
- Price
- Available Offers
- Seller Information
- Stock Availability

This organized catalog helps customers make informed purchasing decisions.

### **3. Smart Search and Filtering**

Amazon's intelligent search engine enables customers to locate products quickly using keywords.

Customers can refine search results using filters such as:

- Brand
- Price Range
- Customer Ratings
- Discount
- Delivery Speed
- Prime Eligibility
- Product Category
- Color
- Size

Machine Learning algorithms also recommend relevant products based on browsing history and previous purchases.

## **4. Shopping Cart**

The Shopping Cart enables customers to add multiple products before completing their purchase.

Features include:

- Update product quantity
- Remove products
- Save items for later
- Estimate shipping charges
- Apply promotional coupons
- Calculate total order value

This feature improves convenience by allowing customers to purchase multiple items in a single transaction.

## **5. Wishlist**

Customers can save products they intend to purchase later using the Wishlist feature.

Benefits include:

- Save favorite products
- Monitor price changes
- Share wishlists with friends and family
- Organize products for future purchases

This increases customer engagement and encourages repeat visits.

## **6. Order Management System**

Once an order is placed, Amazon automatically manages the complete order lifecycle.

Customers can:

- View order history
- Track order status
- Cancel eligible orders
- Download invoices
- Request returns
- Request replacements

This provides complete transparency throughout the purchasing process.

## **7. Secure Payment System**

Amazon supports multiple payment methods to provide flexibility and security.

Available payment options include:

- Credit Card
- Debit Card
- UPI
- Net Banking
- Digital Wallets
- EMI
- Amazon Pay Balance
- Cash on Delivery (COD)
- Gift Cards

Advanced encryption and fraud detection systems ensure secure online transactions.

## **8. Real-Time Order Tracking**

Customers receive live updates after placing an order.

Tracking information includes:

- Order Confirmed
- Packed
- Shipped
- Out for Delivery
- Delivered

Customers receive notifications via email, SMS, and the Amazon mobile application.

## **9. Customer Reviews and Ratings**

Customers can share their shopping experience by providing ratings and written reviews.

Review system includes:

- Star Ratings (1–5)

- Written Reviews
- Customer Images
- Customer Videos
- Verified Purchase Badge

These reviews help future customers evaluate products before purchasing.

## **10. Seller Marketplace**

Amazon allows businesses to sell products directly through its marketplace.

Seller tools include:

- Product Listing
- Inventory Management
- Sales Reports
- Advertising Dashboard
- Order Processing
- Customer Communication

This feature enables businesses of every size to reach millions of customers worldwide.

## **11. Inventory Management**

Amazon continuously monitors inventory across warehouses.

Inventory features include:

- Stock Monitoring
- Automatic Stock Updates
- Low Stock Alerts
- Warehouse Allocation
- Fulfillment Center Management

This minimizes stock shortages and improves order fulfillment.

## **12. Recommendation System**

Amazon uses Artificial Intelligence and Machine Learning to recommend products.

Recommendations are based on:

- Browsing History
- Purchase History
- Recently Viewed Products
- Customer Interests
- Similar Customer Purchases

This personalized shopping experience increases customer satisfaction and sales.

### **13. Return and Refund Management**

Amazon offers a customer-friendly return policy.

Customers can:

- Initiate returns online
- Schedule pickup
- Replace products
- Receive refunds
- Track return status

This reduces customer risk and builds long-term trust.

### **14. Amazon Prime Membership**

Amazon Prime provides premium services through a subscription model.

Prime benefits include:

- Free Fast Delivery
- Same-Day Delivery (selected locations)
- Prime Video
- Prime Music
- Exclusive Deals
- Early Access to Sales

Prime membership improves customer loyalty and increases purchase frequency.

### **15. Customer Support**

Amazon provides customer support through multiple channels.

Support options include:

- Live Chat
- Phone Support
- Email Support
- Help Center
- AI Chatbot

Quick issue resolution enhances customer satisfaction.

# Summary of Top Features

| Feature               | Purpose                                  |
|-----------------------|------------------------------------------|
| Customer Account      | Manages user information and preferences |
| Product Catalog       | Stores and displays products             |
| Smart Search          | Quickly finds products using filters     |
| Shopping Cart         | Collects products before checkout        |
| Wishlist              | Saves products for future purchase       |
| Order Management      | Handles complete order lifecycle         |
| Secure Payment        | Supports multiple secure payment methods |
| Order Tracking        | Tracks delivery status in real time      |
| Reviews & Ratings     | Helps customers evaluate products        |
| Seller Marketplace    | Allows businesses to sell products       |
| Inventory Management  | Monitors product availability            |
| Recommendation System | Suggests personalized products           |
| Return & Refund       | Simplifies product returns               |
| Amazon Prime          | Offers premium shopping benefits         |
| Customer Support      | Resolves customer issues                 |

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# Database Schema Design

## Schema Description

Amazon's e-commerce platform manages millions of customers, products, sellers, orders, payments, and shipments every day. To support these operations efficiently, Amazon requires a well-structured relational database schema. The schema organizes data into multiple interconnected entities, ensuring data consistency, minimizing redundancy, and enabling fast retrieval of information.

The primary entities in the Amazon E-commerce schema include **Customer, Seller, Category, Product, Inventory, Shopping Cart, Cart Items, Orders, Order Items, Payment, Shipment, Reviews, Wishlist, Coupons, and Returns & Refunds**. Each entity contains specific attributes that describe its properties, while relationships between entities

represent real-world interactions such as customers placing orders, sellers listing products, and payments being processed.

## Entity Descriptions

### 1. Customer Entity

The Customer entity stores information about registered users who purchase products from Amazon.

#### Attributes

- **CustomerID (Primary Key):** Unique identifier for each customer.
- **FirstName:** Customer's first name.
- **LastName:** Customer's last name.
- **Email:** Registered email address.
- **PhoneNumber:** Contact number.
- **PasswordHash:** Encrypted password.
- **DateOfBirth:** Customer's date of birth.
- **RegistrationDate:** Date the account was created.
- **PrimeMember:** Indicates whether the customer has an Amazon Prime subscription.
- **AccountStatus:** Active, Suspended, or Closed.

### 2. Seller Entity

The Seller entity stores information about businesses or individuals selling products on Amazon.

#### Attributes

- **SellerID (Primary Key):** Unique seller identifier.
- **BusinessName:** Registered business name.
- **OwnerName:** Name of the business owner.
- **Email**
- **PhoneNumber**
- **GSTNumber**
- **BusinessAddress**
- **RegistrationDate**
- **SellerRating**
- **SellerStatus**

### 3. Category Entity

Categories organize products into different groups for easy browsing.

#### Attributes

- **CategoryID (Primary Key)**
- **CategoryName**
- **Description**
- **ParentCategoryID (Foreign Key)** *(for subcategories)*

Example:

- Electronics
- Fashion
- Books
- Grocery
- Home & Kitchen

## 4. Product Entity

The Product entity stores detailed information about every product available on Amazon.

### Attributes

- **ProductID (Primary Key)**
- **SellerID (Foreign Key → Seller)**
- **CategoryID (Foreign Key → Category)**
- **ProductName**
- **Brand**
- **Description**
- **Price**
- **DiscountPercentage**
- **Weight**
- **Color**
- **Size**
- **Warranty**
- **ImageURL**
- **AverageRating**
- **CreatedDate**

## 5. Inventory Entity

The Inventory entity manages product stock across warehouses.

### Attributes

- **InventoryID (Primary Key)**
- **ProductID (Foreign Key → Product)**
- **WarehouseLocation**
- **AvailableStock**
- **ReservedStock**
- **LastUpdated**

This entity ensures customers only purchase products currently available in stock.

## 6. Shopping Cart Entity

The Shopping Cart stores products selected by customers before checkout.

### Attributes

- **CartID (Primary Key)**
- **CustomerID (Foreign Key → Customer)**
- **CreatedDate**
- **LastModifiedDate**
- **CartStatus**

## 7. Cart Item Entity

A shopping cart can contain multiple products.

### Attributes

- **CartItemID (Primary Key)**
- **CartID (Foreign Key → Shopping Cart)**
- **ProductID (Foreign Key → Product)**
- **Quantity**
- **PriceAtAddition**

## 8. Order Entity

The Order entity stores completed customer purchases.

### Attributes

- **OrderID (Primary Key)**
- **CustomerID (Foreign Key → Customer)**
- **OrderDate**
- **OrderStatus**
- **ShippingAddress**
- **BillingAddress**
- **TotalAmount**
- **CouponID (Foreign Key → Coupon) (Optional)**

Possible Order Status values:

- Pending
- Confirmed
- Packed
- Shipped
- Delivered

- Cancelled

## 9. Order Item Entity

Each order can contain multiple products.

### Attributes

- **OrderItemID (Primary Key)**
- **OrderID (Foreign Key → Order)**
- **ProductID (Foreign Key → Product)**
- **Quantity**
- **UnitPrice**
- **DiscountApplied**
- **TotalPrice**

This entity resolves the many-to-many relationship between Orders and Products.

## 10. Payment Entity

The Payment entity records customer payment details.

### Attributes

- **PaymentID (Primary Key)**
- **OrderID (Foreign Key → Order)**
- **PaymentMethod**
- **PaymentDate**
- **TransactionID**
- **PaymentStatus**
- **AmountPaid**

Supported payment methods include:

- Credit Card
- Debit Card
- UPI
- Net Banking
- Wallet
- EMI
- Cash on Delivery

## 11. Shipment Entity

The Shipment entity manages product delivery.

### Attributes

- **ShipmentID (Primary Key)**
- **OrderID (Foreign Key → Order)**
- **CourierPartner**
- **TrackingNumber**
- **ShipmentDate**
- **ExpectedDeliveryDate**
- **ActualDeliveryDate**
- **ShipmentStatus**

## **12. Review Entity**

Customers can provide feedback after purchasing products.

### **Attributes**

- **ReviewID (Primary Key)**
- **CustomerID (Foreign Key → Customer)**
- **ProductID (Foreign Key → Product)**
- **Rating**
- **ReviewText**
- **ReviewDate**

## **13. Wishlist Entity**

Customers can save products for future purchases.

### **Attributes**

- **WishlistID (Primary Key)**
- **CustomerID (Foreign Key → Customer)**
- **ProductID (Foreign Key → Product)**
- **DateAdded**

## **14. Coupon Entity**

Coupons provide promotional discounts.

### **Attributes**

- **CouponID (Primary Key)**
- **CouponCode**
- **DiscountType**
- **DiscountValue**
- **MinimumOrderValue**
- **StartDate**
- **ExpiryDate**
- **CouponStatus**

## 15. Return & Refund Entity

This entity manages product returns and refunds.

### Attributes

- **ReturnID (Primary Key)**
- **OrderItemID (Foreign Key → Order Item)**
- **ReturnReason**
- **ReturnRequestDate**
- **PickupDate**
- **RefundAmount**
- **RefundStatus**

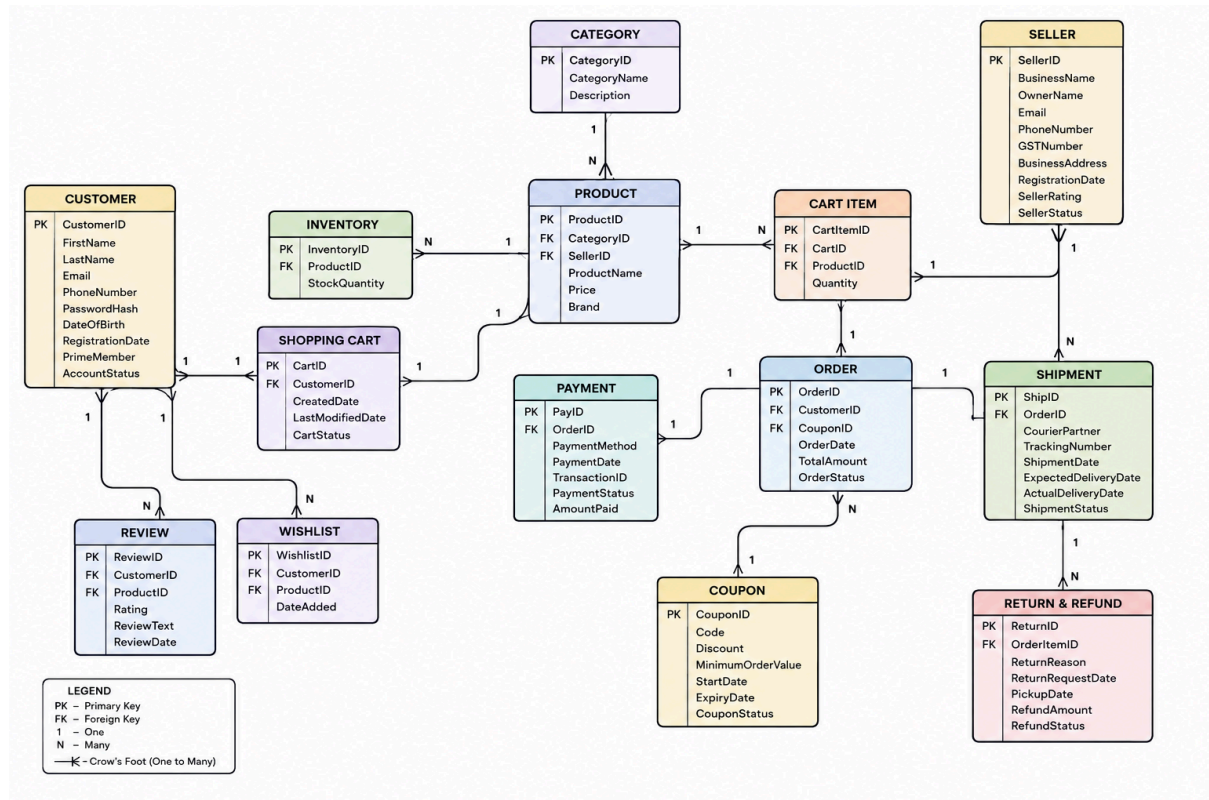
## Relationships Between Entities

The entities are connected through the following relationships:

- **Customer places Orders** — One customer can place many orders, while each order belongs to one customer.
- **Seller lists Products** — One seller can list multiple products, while each product belongs to one seller.
- **Category contains Products** — One category can contain many products, while each product belongs to one category.
- **Product has Inventory** — Each product is associated with inventory records that track stock availability.
- **Customer owns a Shopping Cart** — Each customer has one active shopping cart, which can contain multiple cart items.
- **Shopping Cart contains Products** — A shopping cart can contain many products through the Cart Item entity.
- **Order contains Products** — Each order can include multiple products through the Order Item entity.
- **Order has one Payment** — Every completed order is linked to one payment transaction.
- **Order has one Shipment** — Each order has shipment information used for delivery tracking.
- **Customer writes Reviews** — Customers can review multiple products, and each product can receive reviews from many customers.
- **Customer creates a Wishlist** — Customers can save multiple products to their wishlist.
- **Order uses Coupons** — An order may optionally apply a promotional coupon.
- **Order Items can have Returns** — Each order item may generate a return and refund request if eligible.

# Entity Relationship (ER) Diagram

The following ER Diagram represents the relationships between the major entities of the Amazon E-commerce platform. It illustrates how customers interact with products, sellers manage inventory, orders are processed, payments are completed, and shipments are delivered.



## Cardinality of Relationships

| Relationship              | Cardinality |
|---------------------------|-------------|
| Category → Product        | 1 : Many    |
| Seller → Product          | 1 : Many    |
| Product → Inventory       | 1 : Many    |
| Customer → Shopping Cart  | 1 : 1       |
| Shopping Cart → Cart Item | 1 : Many    |
| Product → Cart Item       | 1 : Many    |
| Customer → Order          | 1 : Many    |

|                              |          |
|------------------------------|----------|
| Order → Order Item           | 1 : Many |
| Product → Order Item         | 1 : Many |
| Order → Payment              | 1 : 1    |
| Order → Shipment             | 1 : 1    |
| Customer → Review            | 1 : Many |
| Product → Review             | 1 : Many |
| Customer → Wishlist          | 1 : Many |
| Product → Wishlist           | 1 : Many |
| Coupon → Order               | 1 : Many |
| Order Item → Return & Refund | 1 : 1    |

## Conclusion

In this case study, the Amazon E-commerce platform demonstrates how an efficient relational database supports every stage of the online shopping experience—from customer registration and product browsing to order placement, payment, shipment, reviews, and returns. The proposed schema reflects real-world business operations by organizing data into well-defined entities connected through primary and foreign keys.

This database design ensures **data integrity**, **high performance**, **scalability**, and **security**, making it suitable for handling millions of transactions and users. It also provides a strong foundation for advanced analytics, customer personalization, inventory optimization, and future enhancements such as AI-driven recommendations and demand forecasting.

Overall, this product dissection highlights the critical role of database schema design in enabling Amazon to deliver a seamless, reliable, and customer-centric e-commerce experience.